

Euler's step method

Basic skills

$$\text{Q1. a) } y^2 \frac{dy}{dx} = 3+x \Rightarrow \frac{dy}{dx} = \frac{3+x}{y^2}$$

$$\text{b) } xy = \sin x \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{xy}{\sin x}$$

$$\text{c) } y = e^x \frac{dy}{dx} - x \frac{dy}{dx} \Rightarrow (e^x - x) \frac{dy}{dx} = y$$

$$\frac{dy}{dx} = \frac{y}{e^x - x}$$

$$\text{d) } y \frac{dy}{dx} = 4 \frac{dy}{dx} - x \frac{dy}{dx} \quad \frac{dy}{dx} = 0$$

$$(\text{or } y = 4 - x)$$

$$\text{Q2. } f(x, y) = 2x^2y + y + 1$$

$$\text{a) } f(1, 1) = 2 \times 1^2 \times 1 + 1 + 1 = 4$$

$$\text{b) } f(0, 1) = 2 \times 0^2 \times 1 + 1 + 1 = 2$$

$$c) f(-1, 1) = f(1, 1) = 4$$

↪ as only x^2 even w.r.t x

$$d) f(3, 7) = 2 \times 3^2 \times 7 + 7 + 1 = 134$$

$$Q3. \quad x_0 = 1 \quad h = 0.5$$

$$a) \quad x_1 = x_0 + h = 1 + 0.5 = 1.5$$

$$b) \quad x_2 = 2 \quad c) \quad x_5 = 3.5$$

↪ $+3h$

$$Q4. \quad y_0 = 2 \quad h = 0.1, \quad f(x_0, y_0) = 20$$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$= 2 + 0.1 \times 20 = 2 + 2 = 4$$

Competency skills

$$Q5. \quad \frac{dy}{dx} = y + 2x^3$$

$$f(x, y) = y + 2x^3$$

$$x_0 = 1, \quad y_0 = y(1) = 3$$

$$h = 0.2$$

$$a) \quad x_1 = x_0 + h = 1 + 0.2 = 1.2$$

$$b) \quad y_1 = y_0 + h f(x_0, y_0) = 3 + 0.2 (3 + 2 \times 1^3) = 4$$

$$c) \quad y_2 = y_1 + h f(x_1, y_1) \\ = 4 + 0.2 (4 + 2 \times 1.2^3) = 5.4912$$

$$d) \quad x_2 = 1.4 \quad x_3 = 1.6 \quad x_4 = 1.8 \quad x_5 = 2$$

$$y_3 = y_2 + h f(x_2, y_2) \\ = 5.4912 + 0.2 (5.4912 + 2 \times 1.4^3) \\ = 7.68784$$

$$y_4 = 7.68784 + 0.2 (7.68784 + 2 \times 1.6^3) \\ = 10.863808$$

$$y_5 = 10.863808 + 0.2 (10.863808 + 2 \times 1.8^3) \\ = 15.3693696 \\ = 15.37 \quad (2 \text{ d.p.})$$

$$e) \quad y(2) = y(x_5) = y_5 = 15.37$$

$$Q6. \quad \frac{dy}{dx} = \underbrace{\cos(x+y) + e^{xy}}_{f(x,y)} \quad y(0)=0 \\ h=0.25$$

$$x_0 = 0 \quad x_1 = 0.25 \quad x_2 = 0.5 \quad x_3 = 0.75$$

$$x_4 = 1$$

$$\text{want } y(1) = y(x_4) = y_4$$

$$\text{so } y_0 = y(0) = 0$$

$$y_1 = y_0 + h f(x_0, y_0)$$

$$= 0 + 0.25 (\cos(0+0) + e^{0 \times 0})$$
$$= 0.5$$

$$y_2 = 0.5 + 0.25 (\cos 0.75 + e^{0.125})$$
$$= 0.966$$

$$y_3 = 0.966 + 0.25 (\cos 1.466 + e^{0.483})$$
$$= 1.397$$

$$y_4 = 1.397 + 0.25 (\cos 2.147 + e^{1.048})$$
$$= 1.974$$

$$\text{so } y(1) = 1.974$$

Q 7. $\sqrt{x^2+y^2} \frac{dy}{dx} = x^2 y^2$ $y(0)=1$ $h=1$
want

so $f(x,y) = \frac{x^2 y^2}{\sqrt{x^2+y^2}}$ $y(3)=y(x_3)=y_3$

$x_0=0$ $x_1=1$ $x_2=2$ $x_3=3$

$y_0=1$

$y_1 = y_0 + h f(x_0, y_0)$
 $= 1 + 1 \times \frac{0^2 \times 1^2}{\sqrt{0^2+1^2}} = 1$

$y_2 = 1 + 1 \times \frac{1^2 \times 1^2}{\sqrt{1^2+1^2}} = 1 + \frac{1}{\sqrt{2}}$

$y_3 = 1 + \frac{1}{\sqrt{2}} + 1 \times \frac{2^2 \times (1+\frac{1}{\sqrt{2}})^2}{\sqrt{2^2+(1+\frac{1}{\sqrt{2}})^2}}$

$= 6.14$

Q8. $\frac{d^2 y}{dx^2} = 2 \frac{dy}{dx} + x^2 + 5$

$\frac{dy}{dx}(1) = y(1) = 1 \quad h = 0.25$

$x_0 = 1 \quad x_1 = 1.25 \quad x_2 = 1.5 \quad x_3 = 1.75$

$x_4 = 2 \quad y(2) = y(x_4) = y_4$

Now let $z = \frac{dy}{dx}$ then ODE is

$f(x, y)$ so $z_n = f(x_n, y_n)$

$\frac{dz}{dx} = 2z + x^2 + 5$ $g(x, y)$

with $z_0 = \frac{dy}{dx}(1) = 1$

need up to z_3 AS IN $f(x_3, y_3) = z_3$

$z_1 = z_0 + h g(x_0, y_0)$

$= 1 + 0.25 \times (2 \times 1 + 1^2 + 5)$

$= 3$

$$z_2 = 3 + 0.25 \times (2 \times 3 + 1.25^2 + 5) \\ = 6.14$$

$$z_3 = 6.14 + 0.25 \times (2 \times 6.14 + 1.5^2 + 5) \\ = 11.02$$

Then

$$y_1 = y_0 + h f(x_0, y_0) \\ = 1 + 0.25 \times z_0 = 1 + 0.25 \times 1 \\ = 1.25$$

$$y_2 = 1.25 + 0.25 \times z_1 \\ = 1.25 + 0.25 \times 3 = 2$$

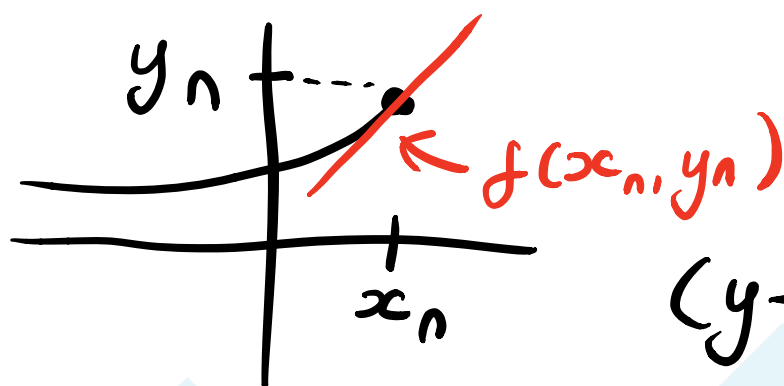
$$y_3 = 3 + 0.25 \times z_2 \\ = 3 + 0.25 \times 6.14 = 4.535$$

$$y_4 = 4.535 + 0.25 \times z_3 \\ = 4.535 + 0.25 \times 11.02 = 7.29$$

For each derivative need to iterate
up to one less place e.g. y_3 needed for y_4

Q9. a)

Assume it's
on the line
with tangent grad



$$(y - y_n) = f(x_n, y_n)(x - x_n)$$

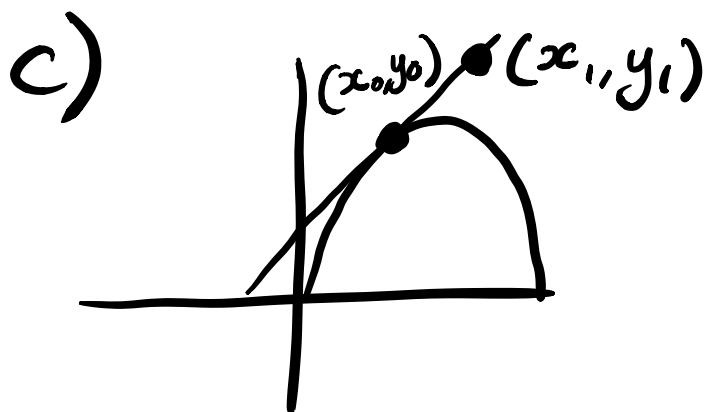
$$\text{so for } x_{n+1} = x_n + h$$

$$y - y_n = f(x_n, y_n)h$$

$$y = y_n + hf(x_n, y_n)$$

so $y_{n+1} = f(x_{n+1})$ is as above

b) using smaller h means travelling
on tangent for less so more
accurate



$$\frac{dy}{dx} = -x$$

$$h = 1 \quad x_0 =$$

Any with non-tangential grad
at next point

Q 10. Let y_{n+1}^* be current est

$$y_{n+1}^* = y_n + h f(x_n, y_n)$$

Then $f(x_{n+1}, y_{n+1}^*)$ is grad at
end of step now retroactively
update our est. so

$$y_{n+1} = y_n + h \frac{1}{2} (f(x_n, y_n) + f(x_{n+1}, y_{n+1}^*))$$

average grad

using average grad alleviates issues

